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Chapter 22 - Inverse Laplace Transforms

Chapter 21 converted time-domain functions into functions of the transform variable s . This chapter reverses that direction: it starts with $F(s)$ and reconstructs a time-domain function $f(t)$.

Write:

$$\mathcal{L}^{-1}\{F(s)\} = f(t).$$

We will learn to:

- recognise inverse transforms from a compact table
- use linearity to separate sums and constant multiples
- complete the square before matching shifted sine and cosine forms
- rewrite numerators around the same shift used in the denominator
- decompose rational functions using partial fractions
- verify a reconstructed function by transforming it forward again

Convolutions and unit-step translations begin in Chapter 23. The examples, figures, derivations, and exercises here are independently constructed; the reference chapter is used only as a topic map.

Reversing The Transform

The Goal Of This Block

The goal is to understand what an inverse Laplace transform claims and how a forward transform can verify the answer.

The Direction Of The Operation

The forward transform takes a function of t and produces a function of s :

$$f(t) \xrightarrow{\mathcal{L}} F(s).$$

The inverse transform travels in the opposite direction:

$$F(s) \xrightarrow{\mathcal{L}^{-1}} f(t).$$

The forward and inverse statements describe the same transform pair:

$$\boxed{\mathcal{L}^{-1}\{F(s)\} = f(t) \quad \Longleftrightarrow \quad \mathcal{L}\{f(t)\} = F(s)}.$$

Here:

- t is the time-domain variable
- s is the transform-domain variable
- $F(s)$ is the expression we are given
- $f(t)$ is the function we want to recover

The superscript -1 means **inverse operation**. It does not mean the reciprocal $1/\mathcal{L}$.

A First Recognition

Chapter 21 established:

$$\mathcal{L}\{e^{-4t}\} = \frac{1}{s+4}.$$

Read the same transform pair in reverse:

$$\mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\} = e^{-4t}.$$

Multiplying the transform by 3 multiplies the time function by 3:

$$\mathcal{L}^{-1}\left\{\frac{3}{s+4}\right\} = 3e^{-4t}.$$

What this tells us:

**inverse transformation is not a new integration trick at every step;
much of the work is recognising a known forward-transform pattern in
reverse.**

Verification By A Forward Transform

Suppose we claim:

$$\mathcal{L}^{-1}\left\{\frac{3}{s+4}\right\} = 3e^{-4t}.$$

To verify the claim $\mathcal{L}^{-1}\{3/(s+4)\} = 3e^{-4t}$, transform the proposed time function forward:

$$\begin{aligned}\mathcal{L}\{3e^{-4t}\} &= 3\mathcal{L}\{e^{-4t}\} \\ &= 3\left(\frac{1}{s+4}\right) \\ &= \frac{3}{s+4}.\end{aligned}$$

The forward transform reproduces the original $F(s)$, so the inverse is correct.

This verification method will remain available even when the reconstruction requires several algebraic steps.

What Denominator Structure Suggests

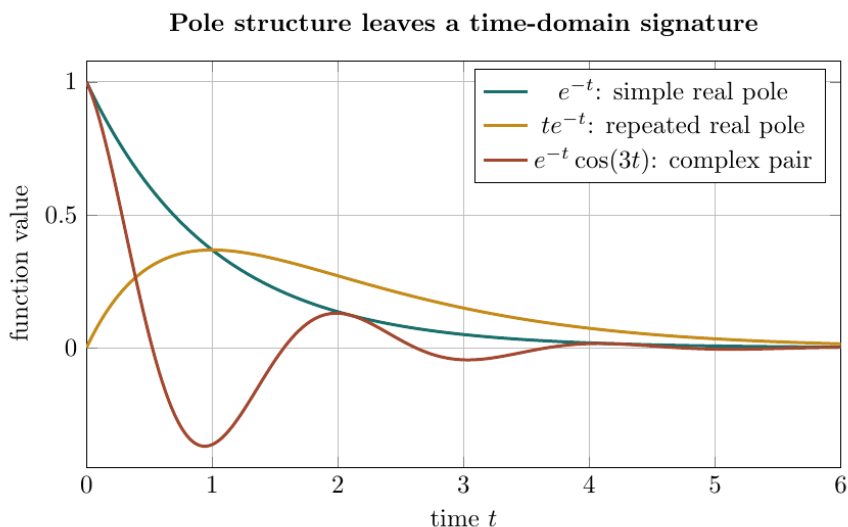


Figure 1: Time-domain signatures of three pole structures

The graph compares three standard time-domain patterns:

- a simple real denominator factor produces an exponential
- a repeated real factor introduces a power of t
- a quadratic associated with a complex pair produces oscillation

The three curves show that denominator structure predicts the **type** of function that should appear, although it does not determine every coefficient.

A useful interpretation:

inspect the denominator first to identify the likely time-domain family; then use the numerator to determine the required mixture and scale.

Block 1 Summary

An inverse Laplace transform reverses a known forward transform. A proposed answer can always be checked by applying $\mathcal{L}$ and confirming that the original expression $F(s)$ returns. For the ordinary piecewise-continuous functions used here, the inverse is unique apart from harmless differences at isolated points.

Direct Recognition And Linearity

The Goal Of This Block

The goal is to read a small inverse-transform table accurately and to separate expressions into manageable terms before matching them.

A Core Inverse Table

The following entries reverse the table developed in Chapter 21:

Transform-domain expression $F(s)$	Time-domain function $f(t)$
$1/s$	1
$1/s^{n+1}$	$t^n/n!$
$1/(s-a)$	e^{at}
$b/(s^2+b^2)$	$\sin(bt)$
$s/(s^2+b^2)$	$\cos(bt)$
$b/(s^2-b^2)$	$\sinh(bt)$
$s/(s^2-b^2)$	$\cosh(bt)$

As a direct table example:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^5}\right\} = \frac{t^4}{4!} = \frac{t^4}{24}.$$

The denominator exponent 5 corresponds to $n + 1 = 5$, so $n = 4$. The factorial is $4!$, not $5!$.

Inverse Linearity

If c_1 and c_2 are constants, then:

$$\boxed{\mathcal{L}^{-1}\{c_1F(s) + c_2G(s)\} = c_1\mathcal{L}^{-1}\{F(s)\} + c_2\mathcal{L}^{-1}\{G(s)\}.$$

Linearity permits two useful actions:

1. split a sum or difference into separate inverse transforms
2. move constant factors outside the inverse-transform symbol

It does **not** permit products to be separated in the same way.

A Complete Direct Example

Find:

$$\mathcal{L}^{-1}\left\{\frac{6}{s^4} - \frac{10}{s^2 + 25} + \frac{3}{s + 2}\right\}.$$

Split the expression by linearity:

$$\begin{aligned} f(t) &= 6\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\} - 10\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 25}\right\} \\ &\quad + 3\mathcal{L}^{-1}\left\{\frac{1}{s + 2}\right\}. \end{aligned}$$

In the direct example, the first term is $6/s^4$. Match s^4 with s^{n+1} . Since $n + 1 = 4$, we have $n = 3$:

$$6\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\} = 6\left(\frac{t^3}{3!}\right) = t^3.$$

For the sine term, write $25 = 5^2$ and manufacture the required numerator 5:

$$\begin{aligned} -\frac{10}{s^2 + 25} &= -2\left(\frac{5}{s^2 + 5^2}\right), \\ -10\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 25}\right\} &= -2\sin(5t). \end{aligned}$$

For the final term, $s + 2 = s - (-2)$:

$$3\mathcal{L}^{-1}\left\{\frac{1}{s + 2}\right\} = 3e^{-2t}.$$

Combine the three reconstructed components:

$$\boxed{f(t) = t^3 - 2\sin(5t) + 3e^{-2t}}.$$

Why The Sine Numerator Matters

The standard pair is:

$$\mathcal{L}\{\sin(bt)\} = \frac{b}{s^2 + b^2}.$$

Solving the sine pair for the inverse of $1/(s^2 + b^2)$ gives:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + b^2} \right\} = \frac{1}{b} \sin(bt),$$

not merely $\sin(bt)$.

The numerator is part of the pattern. Ignoring it changes the amplitude of the reconstructed function.

Common Direct-Recognition Errors

Watch for these mistakes:

- reading $1/(s + 4)$ as e^{4t} instead of e^{-4t}
- using t^n instead of $t^n/n!$ for $1/s^{n+1}$
- confusing the sine numerator b with the cosine numerator s
- trying to invert a product by multiplying two separate inverse transforms

The practical rule:

match the complete fraction, including its numerator, denominator, shifts, and constant factors.

Block 2 Summary

Use linearity first. Then match each term with a complete table entry and adjust any missing numerator constants explicitly.

Completing The Square

The Goal Of This Block

The goal is to rewrite a quadratic denominator in the shifted-square form needed for exponentially weighted sine and cosine functions.

Why The Denominator Comes First

The shifted trigonometric transform pairs are:

$$\begin{aligned}\mathcal{L}\{e^{at} \cos(bt)\} &= \frac{s - a}{(s - a)^2 + b^2}, \\ \mathcal{L}\{e^{at} \sin(bt)\} &= \frac{b}{(s - a)^2 + b^2}.\end{aligned}$$

The denominator identifies both parameters:

- the shift a controls the exponential factor e^{at}
- the positive square b^2 controls the frequency b

Until the denominator has the form $(s - a)^2 + b^2$, the correct numerator patterns are not yet visible.

The Completing-The-Square Pattern

For a monic quadratic:

$$s^2 + ps + q,$$

take half of p , square it, add it inside the square, and compensate in the constant term:

$$\begin{aligned}s^2 + ps + q &= s^2 + ps + \left(\frac{p}{2}\right)^2 + q - \left(\frac{p}{2}\right)^2 \\ &= \left(s + \frac{p}{2}\right)^2 + q - \frac{p^2}{4}.\end{aligned}$$

The sign inside the completed square follows the sign of the linear term. For instance, $s^2 - 6s$ begins with $(s - 3)^2$, not $(s + 3)^2$.

A Constant-Numerator Example

Find:

$$\mathcal{L}^{-1} \left\{ \frac{7}{s^2 - 2s + 10} \right\}.$$

Complete the square in the denominator. Half of -2 is -1 :

$$\begin{aligned} s^2 - 2s + 10 &= (s^2 - 2s + 1) + (10 - 1) \\ &= (s - 1)^2 + 9 \\ &= (s - 1)^2 + 3^2. \end{aligned}$$

The denominator now indicates $a = 1$ and $b = 3$. Rewrite the numerator 7 as $(7/3) \cdot 3$ so that the sine numerator appears:

$$\frac{7}{(s - 1)^2 + 3^2} = \frac{7}{3} \left(\frac{3}{(s - 1)^2 + 3^2} \right).$$

Apply the shifted-sine pair:

$$\boxed{\mathcal{L}^{-1} \left\{ \frac{7}{s^2 - 2s + 10} \right\} = \frac{7}{3} e^t \sin(3t)}.$$

A Numerator That Must Also Be Rewritten

Find:

$$\mathcal{L}^{-1} \left\{ \frac{2s + 11}{s^2 + 6s + 13} \right\}.$$

First complete the square in the denominator. Half of 6 is 3:

$$\begin{aligned} s^2 + 6s + 13 &= (s^2 + 6s + 9) + (13 - 9) \\ &= (s + 3)^2 + 4 \\ &= (s + 3)^2 + 2^2. \end{aligned}$$

Because $s + 3 = s - (-3)$, the time-domain exponential will be e^{-3t} .

The cosine numerator must be $s + 3$, so rewrite $2s + 11$ around that same shift:

$$2s + 11 = 2(s + 3) + 5.$$

Substitute the completed denominator and split the numerator:

$$\begin{aligned} \frac{2s + 11}{s^2 + 6s + 13} &= \frac{2(s + 3) + 5}{(s + 3)^2 + 2^2} \\ &= 2 \frac{s + 3}{(s + 3)^2 + 2^2} + \frac{5}{2} \frac{2}{(s + 3)^2 + 2^2}. \end{aligned}$$

Invert the cosine-shaped and sine-shaped terms separately:

$$\boxed{f(t) = 2e^{-3t} \cos(2t) + \frac{5}{2}e^{-3t} \sin(2t)}.$$

The denominator selected the decay rate and frequency. The rewritten numerator selected the mixture of cosine and sine.

Common Completing-Square Errors

Check each of these points:

- half the coefficient of s before squaring it
- compensate for the square that was added
- identify a from $s - a$, not from the visible sign alone
- rewrite the numerator using exactly the same shifted expression

For example, $(s + 3)^2 = (s - (-3))^2$, so $a = -3$ and the exponential is e^{-3t} .

Block 3 Summary

Complete the denominator first. Then rewrite the numerator as a multiple of $s - a$ plus a constant, and match the resulting cosine and sine terms.

Choosing A Partial-Fraction Form

The Goal Of This Block

The goal is to choose the correct partial-fraction template from the factor structure before solving for any constants.

When Partial Fractions Are Needed

A rational transform has the form:

$$F(s) = \frac{P(s)}{Q(s)},$$

where P and Q are polynomials.

Partial fractions are useful when $Q(s)$ factors into simpler pieces but the whole quotient does not match one table entry.

For $F(s) = P(s)/Q(s)$, this chapter focuses on **proper** rational functions:

$$\deg P < \deg Q.$$

If the numerator degree is at least the denominator degree, polynomial long division comes first. Polynomial terms in s can lead beyond ordinary functions into impulse methods, so those cases are not treated here.

Distinct Linear Factors

For distinct linear factors, assign one constant numerator to each factor:

$$\frac{P(s)}{(s-a)(s-b)} = \frac{A}{s-a} + \frac{B}{s-b}.$$

Each linear denominator then corresponds to an exponential after inversion.

Repeated Linear Factors

If $(s-a)$ occurs to power m , include every power from 1 through m :

$$\boxed{\frac{P(s)}{(s-a)^m} = \frac{A_1}{s-a} + \frac{A_2}{(s-a)^2} + \cdots + \frac{A_m}{(s-a)^m}.$$

For example, a factor $(s+1)^3$ requires:

$$\frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{(s+1)^3}.$$

Skipping an intermediate power makes the decomposition incomplete.

Irreducible Quadratic Factors

For a real quadratic that does not factor into real linear factors, use a linear numerator:

$$\boxed{\frac{P(s)}{s^2 + ps + q} = \frac{Bs + C}{s^2 + ps + q}}.$$

If the quadratic is repeated, include a linear numerator over every power:

$$\frac{B_1s + C_1}{Q(s)} + \frac{B_2s + C_2}{Q(s)^2} + \cdots + \frac{B_ms + C_m}{Q(s)^m}.$$

A constant numerator is not general enough because both sine-shaped and cosine-shaped components may be needed.

A Template-Selection Example

Suppose the denominator is:

$$(s + 2)^2(s^2 + 9).$$

The repeated linear factor contributes two fractions, while the irreducible quadratic contributes one fraction with a linear numerator. Therefore the correct template is:

$$\boxed{\frac{P(s)}{(s + 2)^2(s^2 + 9)} = \frac{A}{s + 2} + \frac{B}{(s + 2)^2} + \frac{Cs + D}{s^2 + 9}}.$$

At this stage, no constants have been calculated. Choosing the form correctly is a separate step from solving for A , B , C , and D .

What Partial Fractions Accomplish

Partial fractions separate a transform into simpler components

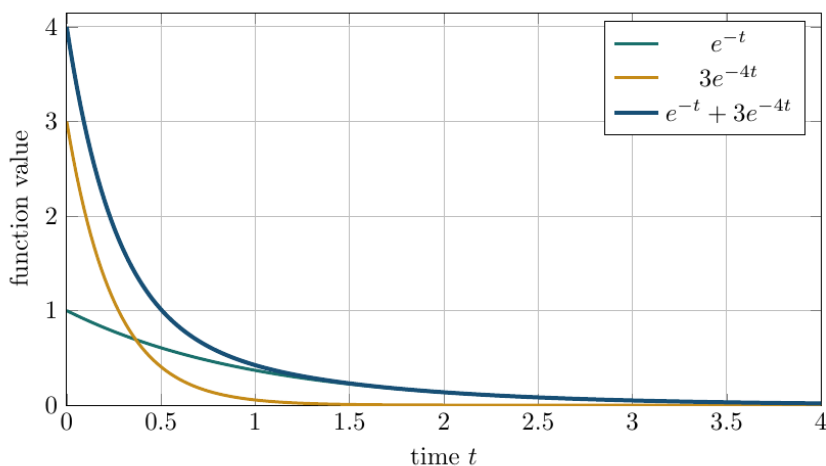


Figure 2: Two partial-fraction components and their reconstructed sum

The graph shows the time-domain functions associated with:

$$\frac{1}{s+1} \quad \text{and} \quad \frac{3}{s+4}.$$

Their inverse transforms are e^{-t} and $3e^{-4t}$. Linearity then rebuilds the inverse of their sum as:

$$e^{-t} + 3e^{-4t}.$$

Partial fractions do not change the rational function. They expose simpler transform pairs that were hidden inside it.

Block 4 Summary

Factor the denominator completely over the real numbers, then assign the correct numerator form to every factor and every repeated power before solving for constants.

Distinct Linear Factors

The Goal Of This Block

The goal is to perform a complete partial-fraction decomposition while making every substituted value and vanished term visible.

A Full Distinct-Factor Example

Find:

$$\mathcal{L}^{-1} \left\{ \frac{4s + 7}{(s + 1)(s + 4)} \right\}.$$

The denominator contains two distinct linear factors, so write:

$$\frac{4s + 7}{(s + 1)(s + 4)} = \frac{A}{s + 1} + \frac{B}{s + 4}.$$

Multiply every term by $(s + 1)(s + 4)$ to clear the denominators:

$$\boxed{4s + 7 = A(s + 4) + B(s + 1)}.$$

This identity must hold for every allowed value of s .

Find A By Choosing S Equals Minus One

Start from the cleared identity:

$$4s + 7 = A(s + 4) + B(s + 1).$$

Set $s = -1$. The factor $s + 1$ becomes zero, so the term containing B vanishes:

$$\begin{aligned} 4(-1) + 7 &= A(-1 + 4) + B(-1 + 1) \\ 3 &= 3A + 0. \end{aligned}$$

Divide by 3:

$$A = 1.$$

Find B By Choosing S Equals Minus Four

Return to the same cleared identity:

$$4s + 7 = A(s + 4) + B(s + 1).$$

Set $s = -4$. The factor $s + 4$ becomes zero, so the term containing A vanishes:

$$\begin{aligned} 4(-4) + 7 &= A(-4 + 4) + B(-4 + 1) \\ -9 &= 0 - 3B. \end{aligned}$$

Divide by -3 :

$$B = 3.$$

Invert The Decomposition

Substitute $A = 1$ and $B = 3$ into the chosen partial-fraction form:

$$\frac{4s + 7}{(s + 1)(s + 4)} = \frac{1}{s + 1} + \frac{3}{s + 4}.$$

Apply inverse linearity and the exponential pair $\mathcal{L}^{-1}\{1/(s - a)\} = e^{at}$:

$$\begin{aligned} f(t) &= \mathcal{L}^{-1}\left\{\frac{1}{s + 1}\right\} + 3\mathcal{L}^{-1}\left\{\frac{1}{s + 4}\right\} \\ &= e^{-t} + 3e^{-4t}. \end{aligned}$$

Substituting the solved decomposition into the inverse transform gives:

$$\boxed{\mathcal{L}^{-1}\left\{\frac{4s + 7}{(s + 1)(s + 4)}\right\} = e^{-t} + 3e^{-4t}.$$

Check The Constants By Recombining

Recombine the two partial fractions over their common denominator:

$$\begin{aligned} \frac{1}{s + 1} + \frac{3}{s + 4} &= \frac{s + 4 + 3(s + 1)}{(s + 1)(s + 4)} \\ &= \frac{s + 4 + 3s + 3}{(s + 1)(s + 4)} \\ &= \frac{4s + 7}{(s + 1)(s + 4)}. \end{aligned}$$

The original transform returns exactly, so both constants are correct.

The Coefficient-Comparison Alternative

The root-substitution method is efficient for distinct linear factors, but coefficient comparison also works.

Expand the cleared identity:

$$\begin{aligned}4s + 7 &= A(s + 4) + B(s + 1) \\ &= (A + B)s + (4A + B).\end{aligned}$$

Match the coefficient of s and the constant term:

$$\begin{aligned}A + B &= 4, \\ 4A + B &= 7.\end{aligned}$$

Subtract the first equation from the second:

$$3A = 3, \quad A = 1.$$

Substitute $A = 1$ into $A + B = 4$:

$$1 + B = 4, \quad B = 3.$$

Both methods produce the same constants.

Block 5 Summary

For distinct linear factors, clear denominators and choose roots that make all but one unknown term vanish. Keep the identity visible during each substitution, then recombine the fractions as a check.

Repeated And Quadratic Factors

The Goal Of This Block

The goal is to handle repeated linear factors and mixed linear-quadratic denominators without omitting required terms.

A Repeated-Linear-Factor Example

Find:

$$\mathcal{L}^{-1} \left\{ \frac{5s^2 + 13s + 6}{(s + 1)^2(s + 3)} \right\}.$$

The repeated factor $(s + 1)^2$ requires both powers 1 and 2:

$$\frac{5s^2 + 13s + 6}{(s + 1)^2(s + 3)} = \frac{A}{s + 1} + \frac{B}{(s + 1)^2} + \frac{C}{s + 3}.$$

Multiply by $(s + 1)^2(s + 3)$:

$$5s^2 + 13s + 6 = A(s + 1)(s + 3) + B(s + 3) + C(s + 1)^2.$$

Find The Repeated-Factor Constants

Set $s = -1$ in the cleared identity. Every term except the one containing B vanishes:

$$\begin{aligned} 5(-1)^2 + 13(-1) + 6 &= A(0)(2) + B(2) + C(0)^2 \\ -2 &= 2B. \end{aligned}$$

Since $-2 = 2B$, dividing both sides by 2 gives:

$$B = -1.$$

Now set $s = -3$. Every term except the one containing C vanishes:

$$\begin{aligned} 5(-3)^2 + 13(-3) + 6 &= A(-2)(0) + B(0) + C(-2)^2 \\ 12 &= 4C. \end{aligned}$$

Dividing $12 = 4C$ by 4 gives:

$$C = 3.$$

To find A , use any convenient remaining value. Set $s = 0$ and substitute $B = -1$ and $C = 3$:

$$\begin{aligned} 6 &= A(1)(3) + (-1)(3) + 3(1)^2 \\ &= 3A - 3 + 3 \\ &= 3A. \end{aligned}$$

Divide by 3:

$$A = 2.$$

Invert The Repeated-Factor Decomposition

Insert $A = 2$, $B = -1$, and $C = 3$:

$$\frac{5s^2 + 13s + 6}{(s+1)^2(s+3)} = \frac{2}{s+1} - \frac{1}{(s+1)^2} + \frac{3}{s+3}.$$

For the decomposition $2/(s+1) - 1/(s+1)^2 + 3/(s+3)$, use the shifted power pairs:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} = e^{-t},$$

For the repeated factor, the second required pair is:

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2} \right\} = te^{-t}.$$

Apply inverse linearity term by term:

$$\boxed{f(t) = 2e^{-t} - te^{-t} + 3e^{-3t}}.$$

The repeated denominator factor produces the extra factor t in the middle term.

A Mixed Linear-Quadratic Example

Find:

$$\mathcal{L}^{-1} \left\{ \frac{14-s}{(s+2)(s^2+4)} \right\}.$$

The linear factor needs a constant numerator. The irreducible quadratic needs a linear numerator:

$$\frac{14-s}{(s+2)(s^2+4)} = \frac{A}{s+2} + \frac{Bs+C}{s^2+4}.$$

Multiplying the mixed decomposition by $(s + 2)(s^2 + 4)$ gives the cleared identity:

$$14 - s = A(s^2 + 4) + (Bs + C)(s + 2).$$

Solve The Mixed Constants

Set $s = -2$ in the cleared identity. The term containing $Bs + C$ vanishes:

$$\begin{aligned} 14 - (-2) &= A((-2)^2 + 4) + (B(-2) + C)(0) \\ 16 &= 8A. \end{aligned}$$

Dividing $16 = 8A$ by 8 gives:

$$A = 2.$$

Substitute $A = 2$ and expand the right side:

$$\begin{aligned} 14 - s &= 2(s^2 + 4) + (Bs + C)(s + 2) \\ &= 2s^2 + 8 + Bs^2 + (2B + C)s + 2C \\ &= (2 + B)s^2 + (2B + C)s + (8 + 2C). \end{aligned}$$

The left side has coefficient 0 on s^2 , coefficient -1 on s , and constant term 14. Match these coefficients:

$$\begin{aligned} 2 + B &= 0, \\ 2B + C &= -1, \\ 8 + 2C &= 14. \end{aligned}$$

The coefficient equations include $2 + B = 0$ and $2B + C = -1$. The first equation gives:

$$B = -2.$$

Substitute $B = -2$ into the second equation:

$$2(-2) + C = -1, \quad C = 3.$$

The constant equation confirms the result because $8 + 2(3) = 14$.

Invert The Mixed Decomposition

Substitute $A = 2$, $B = -2$, and $C = 3$:

$$\frac{14 - s}{(s + 2)(s^2 + 4)} = \frac{2}{s + 2} - \frac{2s}{s^2 + 4} + \frac{3}{s^2 + 4}.$$

Rewrite the final numerator to expose the sine entry:

$$\frac{3}{s^2 + 4} = \frac{3}{2} \frac{2}{s^2 + 2^2}.$$

Invert all three terms:

$$f(t) = 2e^{-2t} - 2\cos(2t) + \frac{3}{2}\sin(2t).$$

Why Root Substitution Was Not Enough

In the mixed example, the real root $s = -2$ isolated A . The quadratic $s^2 + 4$ has no real roots, so real root substitution could not isolate B and C .

Coefficient comparison supplied the remaining equations. This is why the most reliable general strategy is:

1. use convenient roots where they help
2. expand and compare coefficients for what remains

Block 6 Summary

Repeated linear factors require every denominator power. Irreducible quadratics require linear numerators. Root substitution and coefficient comparison can be combined in the same decomposition.

A Reliable End-To-End Workflow

The Goal Of This Block

The goal is to combine factoring, partial fractions, completing the square, numerator matching, inversion, and verification in one uninterrupted example.

The Workflow

For a proper rational transform $F(s) = P(s)/Q(s)$:

1. factor $Q(s)$ over the real numbers
2. choose the complete partial-fraction template
3. clear denominators
4. solve the constants using roots and coefficient comparison
5. complete squares in any quadratic denominators
6. rewrite each numerator around the denominator's shift
7. apply inverse linearity and table entries
8. transform the answer forward to verify it

Do not complete every possible algebraic operation at once. Each stage has a different purpose.

An End-To-End Example

Find:

$$\mathcal{L}^{-1} \left\{ \frac{4s + 18}{(s + 2)(s^2 + 2s + 10)} \right\}.$$

The quadratic discriminant is:

$$\Delta = 2^2 - 4(1)(10) = -36 < 0,$$

Because $\Delta = -36 < 0$, the quadratic is irreducible over the real numbers.
Choose:

$$\frac{4s + 18}{(s + 2)(s^2 + 2s + 10)} = \frac{A}{s + 2} + \frac{Bs + C}{s^2 + 2s + 10}.$$

Clear denominators:

$$4s + 18 = A(s^2 + 2s + 10) + (Bs + C)(s + 2).$$

Solve The Constants Locally

Set $s = -2$ in the cleared identity:

$$\begin{aligned} 4(-2) + 18 &= A((-2)^2 + 2(-2) + 10) + (B(-2) + C)(0) \\ 10 &= 10A. \end{aligned}$$

Dividing $10 = 10A$ by 10 gives:

$$A = 1.$$

For the end-to-end example, root substitution gave $A = 1$. Substitute this value and expand:

$$\begin{aligned} 4s + 18 &= s^2 + 2s + 10 + (Bs + C)(s + 2) \\ &= s^2 + 2s + 10 + Bs^2 + (2B + C)s + 2C \\ &= (1 + B)s^2 + (2 + 2B + C)s + (10 + 2C). \end{aligned}$$

Match the s^2 coefficient:

$$1 + B = 0, \quad B = -1.$$

In the expanded identity, the s -coefficient equation is $2 + 2B + C = 4$. Substitute $B = -1$:

$$\begin{aligned} 2 + 2B + C &= 4, \\ 2 + 2(-1) + C &= 4, \\ C &= 4. \end{aligned}$$

The constant term verifies the result:

$$10 + 2C = 10 + 2(4) = 18.$$

Prepare The Quadratic Term

Insert $A = 1$, $B = -1$, and $C = 4$:

$$\frac{4s + 18}{(s + 2)(s^2 + 2s + 10)} = \frac{1}{s + 2} + \frac{-s + 4}{s^2 + 2s + 10}.$$

For the remaining denominator $s^2 + 2s + 10$, complete the square:

$$\begin{aligned} s^2 + 2s + 10 &= (s^2 + 2s + 1) + (10 - 1) \\ &= (s + 1)^2 + 9 \\ &= (s + 1)^2 + 3^2. \end{aligned}$$

Rewrite the numerator around $s + 1$:

$$-s + 4 = -(s + 1) + 5.$$

Substitute and separate the cosine and sine patterns:

$$\begin{aligned} \frac{-s + 4}{s^2 + 2s + 10} &= -\frac{s + 1}{(s + 1)^2 + 3^2} + \frac{5}{(s + 1)^2 + 3^2} \\ &= -\frac{s + 1}{(s + 1)^2 + 3^2} + \frac{5}{3} \frac{3}{(s + 1)^2 + 3^2}. \end{aligned}$$

Reconstruct The Time Function

Invert the three prepared terms:

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{1}{s + 2} \right\} &= e^{-2t}, \\ \mathcal{L}^{-1} \left\{ -\frac{s + 1}{(s + 1)^2 + 3^2} \right\} &= -e^{-t} \cos(3t), \\ \mathcal{L}^{-1} \left\{ \frac{5}{3} \frac{3}{(s + 1)^2 + 3^2} \right\} &= \frac{5}{3} e^{-t} \sin(3t). \end{aligned}$$

Combine the components:

$$f(t) = e^{-2t} - e^{-t} \cos(3t) + \frac{5}{3} e^{-t} \sin(3t).$$

Verify The End-To-End Answer

Transform the proposed answer term by term:

$$\begin{aligned}\mathcal{L}\{e^{-2t}\} &= \frac{1}{s+2}, \\ \mathcal{L}\{-e^{-t}\cos(3t)\} &= -\frac{s+1}{(s+1)^2+9}, \\ \mathcal{L}\left\{\frac{5}{3}e^{-t}\sin(3t)\right\} &= \frac{5}{(s+1)^2+9}.\end{aligned}$$

Combine the two terms with the quadratic denominator:

$$-\frac{s+1}{(s+1)^2+9} + \frac{5}{(s+1)^2+9} = \frac{-s+4}{s^2+2s+10}.$$

To finish the verification, recombine the two prepared fractions:

$$\frac{1}{s+2} + \frac{-s+4}{s^2+2s+10} = \frac{4s+18}{(s+2)(s^2+2s+10)}.$$

The forward transform reproduces the original expression, completing the verification.

Fast Diagnostic Checks

Before accepting an inverse transform, ask:

- **Properness:** is the numerator degree below the denominator degree?
- **Factors:** has the denominator been factored completely over the reals?
- **Template:** is every repeated power present?
- **Quadratics:** does each irreducible quadratic have a linear numerator?
- **Shift:** does $(s-a)$ produce e^{at} with the correct sign?
- **Frequency:** was b taken from b^2 rather than from the square directly?
- **Amplitude:** was the required sine numerator manufactured explicitly?
- **Verification:** does the forward transform reconstruct the original $F(s)$?

Block 7 Summary

The reliable order is denominator structure, decomposition, constant solving, quadratic preparation, numerator matching, inversion, and forward verification. Keeping those stages separate prevents most sign and amplitude errors.

Original Practice Set

Problems 1 To 4: Direct Recognition

Problem 1: Reverse A Power Transform. Find:

$$\mathcal{L}^{-1} \left\{ \frac{8}{s^5} \right\}.$$

Show how the denominator exponent determines the power of t .

Problem 2: Match A Sine Transform. Find:

$$\mathcal{L}^{-1} \left\{ \frac{4}{s^2 + 16} \right\}.$$

Problem 3: Match A Cosine Transform. Find:

$$\mathcal{L}^{-1} \left\{ \frac{3s}{s^2 + 9} \right\}.$$

Problem 4: Use Inverse Linearity. Find:

$$\mathcal{L}^{-1} \left\{ \frac{2}{s} - \frac{12}{s^3} + \frac{6}{s^2 + 9} \right\}.$$

Problems 5 To 8: Squares And Distinct Factors

Problem 5: Complete A Square. Find:

$$\mathcal{L}^{-1} \left\{ \frac{6}{s^2 - 4s + 13} \right\}.$$

Problem 6: Rewrite A Shifted Numerator. Find:

$$\mathcal{L}^{-1} \left\{ \frac{s + 7}{s^2 + 4s + 20} \right\}.$$

Problem 7: Decompose Distinct Decay Factors. Find:

$$\mathcal{L}^{-1} \left\{ \frac{3(s + 6)}{(s + 2)(s + 5)} \right\}.$$

Problem 8: Decompose A Growing And A Decaying Mode. Find:

$$\mathcal{L}^{-1} \left\{ \frac{2s + 1}{(s - 1)(s + 2)} \right\}.$$

Problems 9 To 12: Repeated And Mixed Factors

Problem 9: Use A Repeated Linear Factor. Find:

$$\mathcal{L}^{-1} \left\{ \frac{s^2 + 2s + 10}{(s + 1)^2(s + 4)} \right\}.$$

Problem 10: Combine A Linear And A Quadratic Factor. Find:

$$\mathcal{L}^{-1} \left\{ \frac{15 - s}{(s + 3)(s^2 + 9)} \right\}.$$

Problem 11: Diagnose An Incomplete Template. A learner proposes:

$$\frac{P(s)}{(s+2)^2(s+5)} = \frac{A}{(s+2)^2} + \frac{B}{s+5}.$$

Explain what is missing and write the complete partial-fraction template.

Problem 12: Complete The Full Workflow. Find and verify:

$$\mathcal{L}^{-1} \left\{ \frac{11s + 27}{(s+1)(s^2 + 6s + 13)} \right\}.$$

Worked Answers: Problems 1 To 4

Answer 1. Match s^5 with s^{n+1} . Solve for n :

$$n + 1 = 5, \quad n = 4.$$

Use $\mathcal{L}^{-1}\{1/s^{n+1}\} = t^n/n!$:

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{8}{s^5}\right\} &= 8\frac{t^4}{4!} \\ &= 8\frac{t^4}{24} \\ &= \boxed{\frac{t^4}{3}}.\end{aligned}$$

Answer 2. Write $16 = 4^2$. The numerator already equals the required frequency 4:

$$\frac{4}{s^2 + 16} = \frac{4}{s^2 + 4^2}.$$

Apply the sine pair:

$$\boxed{\mathcal{L}^{-1}\left\{\frac{4}{s^2 + 16}\right\} = \sin(4t)}.$$

Answer 3. Write $9 = 3^2$ and move the constant 3 outside:

$$\frac{3s}{s^2 + 9} = 3\frac{s}{s^2 + 3^2}.$$

For Answer 3, the prepared term is $3s/(s^2 + 3^2)$. Applying the cosine pair gives:

$$\mathcal{L}^{-1} \left\{ \frac{3s}{s^2 + 9} \right\} = 3 \cos(3t).$$

Answer 4. Apply inverse linearity:

$$f(t) = 2\mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - 12\mathcal{L}^{-1} \left\{ \frac{1}{s^3} \right\} + 6\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 9} \right\}.$$

Evaluate each term using the core table:

$$\begin{aligned} 2\mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} &= 2, \\ -12\mathcal{L}^{-1} \left\{ \frac{1}{s^3} \right\} &= -12 \frac{t^2}{2!} = -6t^2, \\ 6\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 9} \right\} &= 2\mathcal{L}^{-1} \left\{ \frac{3}{s^2 + 3^2} \right\} = 2 \sin(3t). \end{aligned}$$

Combining the three independently inverted terms gives:

$$f(t) = 2 - 6t^2 + 2 \sin(3t).$$

Worked Answers: Problems 5 To 8**Answer 5.** Complete the square:

$$\begin{aligned}
 s^2 - 4s + 13 &= (s^2 - 4s + 4) + (13 - 4) \\
 &= (s - 2)^2 + 9 \\
 &= (s - 2)^2 + 3^2.
 \end{aligned}$$

Rewrite $6 = 2 \cdot 3$ to produce the sine numerator:

$$\frac{6}{(s - 2)^2 + 3^2} = 2 \frac{3}{(s - 2)^2 + 3^2}.$$

Use the shifted-sine pair:

$$\mathcal{L}^{-1} \left\{ \frac{6}{s^2 - 4s + 13} \right\} = 2e^{2t} \sin(3t).$$

Answer 6. Complete the denominator square:

$$\begin{aligned}
 s^2 + 4s + 20 &= (s^2 + 4s + 4) + (20 - 4) \\
 &= (s + 2)^2 + 16 \\
 &= (s + 2)^2 + 4^2.
 \end{aligned}$$

Rewrite the numerator using the same shift:

$$s + 7 = (s + 2) + 5.$$

Split the fraction and manufacture the sine numerator 4:

$$\begin{aligned}
 \frac{s + 7}{s^2 + 4s + 20} &= \frac{s + 2}{(s + 2)^2 + 4^2} + \frac{5}{(s + 2)^2 + 4^2} \\
 &= \frac{s + 2}{(s + 2)^2 + 4^2} + \frac{5}{4} \frac{4}{(s + 2)^2 + 4^2}.
 \end{aligned}$$

For Answer 6, the prepared terms are:

$$\frac{s+2}{(s+2)^2+4^2} + \frac{5}{4} \frac{4}{(s+2)^2+4^2}.$$

Invert these two terms:

$$f(t) = e^{-2t} \cos(4t) + \frac{5}{4} e^{-2t} \sin(4t).$$

Answer 7. Choose the distinct-linear-factor form:

$$\frac{3(s+6)}{(s+2)(s+5)} = \frac{A}{s+2} + \frac{B}{s+5}.$$

Clear denominators:

$$3(s+6) = A(s+5) + B(s+2).$$

Set $s = -2$. The term containing B vanishes:

$$3(-2+6) = A(-2+5), \quad 12 = 3A, \quad A = 4.$$

Set $s = -5$. The term containing A vanishes:

$$3(-5+6) = B(-5+2), \quad 3 = -3B, \quad B = -1.$$

Substitute the constants and invert:

$$\begin{aligned} \frac{3(s+6)}{(s+2)(s+5)} &= \frac{4}{s+2} - \frac{1}{s+5}, \\ f(t) &= \boxed{4e^{-2t} - e^{-5t}}. \end{aligned}$$

Answer 8. Use:

$$\frac{2s+1}{(s-1)(s+2)} = \frac{A}{s-1} + \frac{B}{s+2}.$$

Clear denominators:

$$2s+1 = A(s+2) + B(s-1).$$

Set $s = 1$ to isolate A :

$$2(1)+1 = A(1+2), \quad 3 = 3A, \quad A = 1.$$

Set $s = -2$ to isolate B :

$$2(-2)+1 = B(-2-1), \quad -3 = -3B, \quad B = 1.$$

The decomposition and inverse are:

$$\frac{2s+1}{(s-1)(s+2)} = \frac{1}{s-1} + \frac{1}{s+2},$$
$$f(t) = \boxed{e^t + e^{-2t}}.$$

Worked Answers: Problems 9 To 12

Answer 9. The repeated factor requires both $(s + 1)^{-1}$ and $(s + 1)^{-2}$:

$$\frac{s^2 + 2s + 10}{(s + 1)^2(s + 4)} = \frac{A}{s + 1} + \frac{B}{(s + 1)^2} + \frac{C}{s + 4}.$$

Clear denominators:

$$s^2 + 2s + 10 = A(s + 1)(s + 4) + B(s + 4) + C(s + 1)^2.$$

Set $s = -1$ to isolate B :

$$\begin{aligned}(-1)^2 + 2(-1) + 10 &= B(-1 + 4), \\ 9 &= 3B, \\ B &= 3.\end{aligned}$$

Set $s = -4$ to isolate C :

$$\begin{aligned}(-4)^2 + 2(-4) + 10 &= C(-4 + 1)^2, \\ 18 &= 9C, \\ C &= 2.\end{aligned}$$

Set $s = 0$ and substitute $B = 3$ and $C = 2$:

$$\begin{aligned}10 &= A(1)(4) + 3(4) + 2(1)^2 \\ &= 4A + 14.\end{aligned}$$

Subtract 14 and divide by 4:

$$A = -1.$$

For Answer 9, $A = -1$, $B = 3$, and $C = 2$. Substitute these constants into the partial-fraction form:

$$\frac{s^2 + 2s + 10}{(s + 1)^2(s + 4)} = -\frac{1}{s + 1} + \frac{3}{(s + 1)^2} + \frac{2}{s + 4}.$$

Invert the three terms:

$$\boxed{f(t) = -e^{-t} + 3te^{-t} + 2e^{-4t}}.$$

Answer 10. Choose:

$$\frac{15 - s}{(s + 3)(s^2 + 9)} = \frac{A}{s + 3} + \frac{Bs + C}{s^2 + 9}.$$

Clear denominators:

$$15 - s = A(s^2 + 9) + (Bs + C)(s + 3).$$

Set $s = -3$ to isolate A :

$$15 - (-3) = A((-3)^2 + 9), \quad 18 = 18A, \quad A = 1.$$

Substitute $A = 1$ and expand:

$$\begin{aligned} 15 - s &= s^2 + 9 + (Bs + C)(s + 3) \\ &= (1 + B)s^2 + (3B + C)s + (9 + 3C). \end{aligned}$$

For Answer 10, the expanded identity has s^2 coefficient $1 + B$. Match it with the zero s^2 coefficient on the left:

$$1 + B = 0, \quad B = -1.$$

Match the s coefficient:

$$3(-1) + C = -1, \quad C = 2.$$

For Answer 10, the solved constants are $A = 1$, $B = -1$, and $C = 2$. The constant check is $9 + 3(2) = 15$. Substituting the constants gives:

$$\frac{15 - s}{(s + 3)(s^2 + 9)} = \frac{1}{s + 3} - \frac{s}{s^2 + 9} + \frac{2}{s^2 + 9},$$

$$f(t) = \boxed{e^{-3t} - \cos(3t) + \frac{2}{3} \sin(3t)}.$$

Answer 11. The factor $(s+2)^2$ is repeated. A complete decomposition must include both the first and second powers of $s + 2$.

The missing term is a constant over $s + 2$. The correct template is:

$$\boxed{\frac{P(s)}{(s + 2)^2(s + 5)} = \frac{A}{s + 2} + \frac{B}{(s + 2)^2} + \frac{C}{s + 5}}.$$

Using different constant names is harmless; including every required power is essential.

Answer 12. Choose the linear-quadratic template:

$$\frac{11s + 27}{(s + 1)(s^2 + 6s + 13)} = \frac{A}{s + 1} + \frac{Bs + C}{s^2 + 6s + 13}.$$

For Answer 12, multiply the chosen decomposition by $(s + 1)(s^2 + 6s + 13)$ to clear the denominators:

$$\boxed{11s + 27 = A(s^2 + 6s + 13) + (Bs + C)(s + 1)}.$$

Set $s = -1$ to isolate A :

$$11(-1) + 27 = A((-1)^2 + 6(-1) + 13)$$

$$16 = 8A,$$

$$A = 2.$$

For Answer 12, root substitution in the cleared identity gives $A = 2$. Substitute this value and expand:

$$\begin{aligned} 11s + 27 &= 2(s^2 + 6s + 13) + (Bs + C)(s + 1) \\ &= (2 + B)s^2 + (12 + B + C)s + (26 + C). \end{aligned}$$

Match the s^2 coefficient:

$$2 + B = 0, \quad B = -2.$$

Match the s coefficient and substitute $B = -2$:

$$12 - 2 + C = 11, \quad C = 1.$$

The constant check is $26 + 1 = 27$. Substituting $A = 2$, $B = -2$, and $C = 1$ into the partial-fraction form gives:

$$\frac{11s + 27}{(s + 1)(s^2 + 6s + 13)} = \frac{2}{s + 1} + \frac{-2s + 1}{s^2 + 6s + 13}.$$

For Answer 12's remaining denominator $s^2 + 6s + 13$, complete the square:

$$\begin{aligned} s^2 + 6s + 13 &= (s + 3)^2 + 4 \\ &= (s + 3)^2 + 2^2. \end{aligned}$$

Rewrite the quadratic numerator around $s + 3$:

$$-2s + 1 = -2(s + 3) + 7.$$

Separate the table patterns:

$$\frac{-2s + 1}{s^2 + 6s + 13} = -2 \frac{s + 3}{(s + 3)^2 + 2^2} + \frac{7}{2} \frac{2}{(s + 3)^2 + 2^2}.$$

Invert these three terms:

$$\boxed{f(t) = 2e^{-t} - 2e^{-3t} \cos(2t) + \frac{7}{2}e^{-3t} \sin(2t)}.$$

Verify by transforming the boxed function forward:

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \frac{2}{s + 1} - 2 \frac{s + 3}{(s + 3)^2 + 4} + \frac{7}{(s + 3)^2 + 4} \\ &= \frac{2}{s + 1} + \frac{-2s + 1}{s^2 + 6s + 13} \\ &= \frac{11s + 27}{(s + 1)(s^2 + 6s + 13)}. \end{aligned}$$

The forward transform returns the given expression.

Chapter 22 Final Summary

The inverse transform reverses a known forward transform:

$$\mathcal{L}^{-1}\{F(s)\} = f(t) \iff \mathcal{L}\{f(t)\} = F(s).$$

For rational transforms, use this order:

1. confirm that the rational function is proper
2. factor the denominator over the real numbers
3. choose a complete partial-fraction template
4. clear denominators and solve the constants locally
5. complete any quadratic squares
6. rewrite numerators around the same shifts
7. invert term by term using linearity
8. verify by transforming the result forward

The factor templates to remember are:

$$(s-a)^m \rightarrow \frac{A_1}{s-a} + \dots + \frac{A_m}{(s-a)^m},$$

$$Q(s)^m \rightarrow \frac{B_1s + C_1}{Q(s)} + \dots + \frac{B_ms + C_m}{Q(s)^m}.$$

The central working principle is:

expose one recognisable transform pair at a time, and make every algebraic change visible before applying the inverse-transform table.